

Homework # 3: Informative and non-informative priors. Utility theory and applications. intro to WinBUGS.

Due February 14

1. Service times of a queuing system follow Exponential distribution with an unknown parameter θ . A sample of service times $X_1, \dots, X_n$ is observed.
 - (a) Show that the $\text{Gamma}(\alpha, \lambda)$ family of prior distributions is conjugate.
 - (b) Find the posterior parameters, posterior mean and variance.
 - (c) Suppose that we can allow $\alpha = 0$ and consider a $\text{Gamma}(0, \lambda)$ prior density $\pi(\theta) = 1/\theta$ for $\theta > 0$. Of course, this is not a proper density. Nevertheless, find the posterior distribution, its mean and variance.

2. (Berger, 3.8) For each of the following densities, state whether it represents a location family or a scale family and give the natural non-informative prior for the unknown parameter.

(a) $\text{Uniform}(\theta - 1, \theta + 1)$,

(b) $\text{Cauchy}(0, \beta)$ with a density $f(x | \beta) = \frac{\beta}{\pi(\beta^2 + x^2)}$, $x \in (-\infty, \infty)$.

(c) $\text{Pareto}(2, \sigma)$ with a density $f(x | 2, \sigma) = 2\sigma^2 x^{-3}$, $x > \sigma$
(it is in our inventory of distributions).

3. (Berger, 4.14) Assume $\mathbf{X}$ is a sample from $\text{Poisson}(\theta)$ distribution. The improper non-informative prior

$$\pi(\theta) = \frac{1}{\theta}, \quad \theta > 0,$$

is to be used.

- (a) Find the posterior density of θ , given $\mathbf{X} = \mathbf{x}$ for $\mathbf{x} \neq (0, \dots, 0)$.
- (b) Show that for $\mathbf{x} = (0, \dots, 0)$, the posterior density does not exist. Therefore, Bayesian inference is impossible with this prior... *and this makes good sense!*

For example, consider estimating the average frequency of traffic accidents on a segment of a highway, and suppose that no accidents occur during the period of observation. How can we estimate the frequency θ in this case?

4. (Berger, 2.3) Determine and sketch your own personal utility function for money over the interval from \$-1000 and \$1000.
5. (Berger, 2.4) Look at the following three pairs of gambles:

The 1st pair: $\frac{1}{4}\langle 250 \rangle + \frac{3}{4}\langle 0 \rangle$ or $\frac{1}{2}\langle 40 \rangle + \frac{1}{2}\langle 70 \rangle$
The 2nd pair: $\frac{1}{2}\langle 400 \rangle + \frac{1}{2}\langle -100 \rangle$ or $\frac{2}{3}\langle 150 \rangle + \frac{1}{3}\langle 0 \rangle$
The 3rd pair: $\frac{1}{2}\langle -1000 \rangle + \frac{1}{2}\langle 1000 \rangle$ or $\frac{1}{2}\langle -50 \rangle + \frac{1}{2}\langle 50 \rangle$

- (a) Explain the meaning of gambles in the first pair. How much can you gain with this gamble, and what are the chances?

- (b) Without referring to your utility function from the previous exercise, decide which gamble you prefer in each of the following pairs.
 - (c) Now find which gamble would be preferred, as determined by your utility function constructed in the previous exercise. Is there any discrepancy between your answers in (b) and (c)? If yes, then you may revise your sketch, but for this homework, it is not required.
6. Use BUGS or JAGS to solve #1 of this homework. Service times of a queuing system follow Exponential distribution with an unknown parameter θ . A service time $X = 1.5$ min is observed.
- (a) Assume a $\text{Gamma}(\alpha, \lambda)$ prior distribution with $\alpha = 4$ and $\lambda = 2$. (Notice that WinBUGS understands the second parameter of `dgamma(alpha, lambda)` as the “rate” parameter λ . The mean of this distribution is α/λ .) Find the posterior mean and variance of θ .
 - (b) For an (almost) non-informative prior, assume $\text{Gamma}(\alpha, \lambda)$ with $\alpha = 0.0001$. Obtain the graph of the posterior distribution, its mean and variance.